

Feel the Breeze 🌬️

The High-Energy, Fan-Powered Guide to Sustainable Kubernetes

David Pech — Kubernetes, ArgoCD, AWS, OCI, Postgres fan

Petr Rais — Software Engineer & DevOps enthusiast

Who



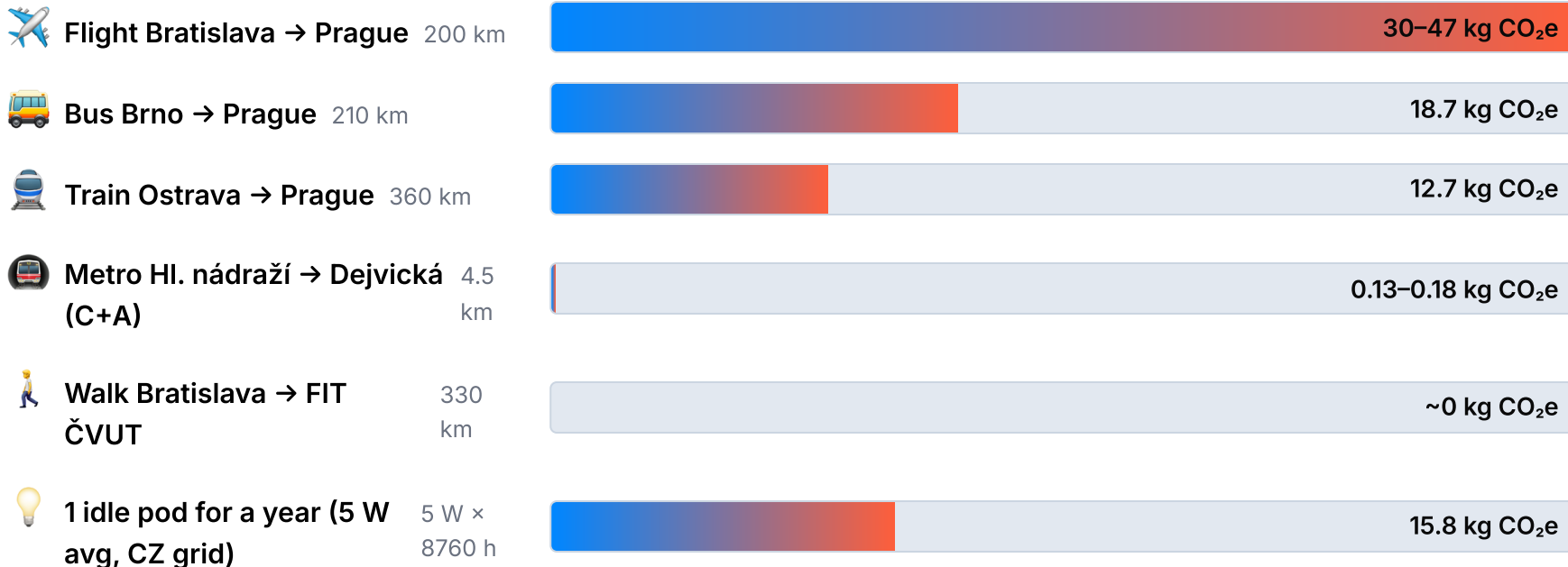
David Pech



Petr Rais



How did you get here? 🚄 ✈️ 🚌



🔥 Burns fuel (jet, diesel)
liters × emission factor

⚡ Uses electricity (metro, train, *your pods*)
kWh × grid intensity (CZ ≈ 0.36 kg/kWh)

💡 One always-on 5 W pod ≈ **one Ostrava→Prague train trip** per year. Multiply by your replica count.

Even hyperscalers hit the wall

Meta cancelled its 200 MW Zeewolde campus (NL) in early 2022:

- 166 ha of farmland
- ~1.4 TWh/yr power demand
- Dutch Senate motion against it
- Local + parliamentary opposition

Netherlands hyperscale moratorium still in force (≥ 70 MW / ≥ 10 ha).

Microsoft froze ~1.5 GW of self-built DCs globally in early 2025 (UK, AU, ND).

Sources: Reuters · [NL Times](#) · [DataCenterDynamics](#) · TD Cowen.

k0s on the Edge: Kubernetes that fits in your pocket 🥧

k0s — single static binary, zero config

- No etcd to manage, no separate binaries
- Runs on **Raspberry Pi 4 · ARM64 · 2 GB RAM**
- `curl | sh` install, systemd service
- Same `kubectl` as anywhere else

Rebuilt from git in minutes:

- Wipe the Pi → reinstall k0s → profit
- ArgoCD re-syncs all workloads automatically
- No manual state to recover

What fits on a Raspberry Pi 4 (2 GB)?

k0s control+worker **~150 MB**

coredns ~20 MB · node-exporter ~15 MB

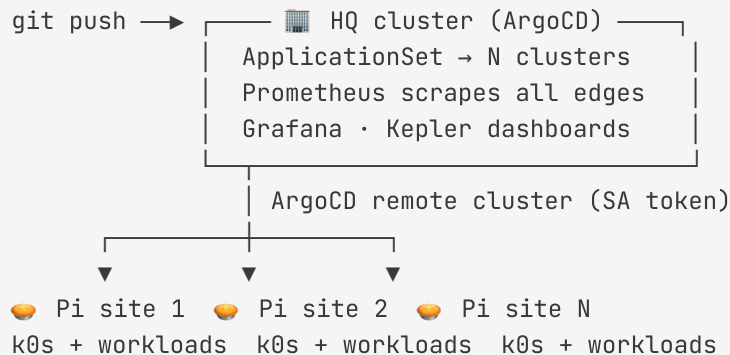
kube-state-metrics ~40 MB · Kepler DS ~60 MB

⇒ **~285 MB observability+control**

⇒ **~1.7 GB free for your workloads**

k0s distro: `k0sproject.io` — CNCF member project, production-grade.

One HQ, N edge clusters — ArgoCD remote management 🏢



ArgoCD remote cluster feature

- Register edge cluster with a ServiceAccount token
- HQ ArgoCD manages apps on the remote k0s node
- No ArgoCD agent on the edge — HQ reaches out
- `ApplicationSet` fans out one app to N clusters

HQ does the heavy lifting

- 1 ArgoCD $\approx$ 250 MB amortized across N sites
- HQ Prometheus scrapes edge via API-server proxy
- HQ can **suspend off-hours** (kube-green, scale-to-zero)
- 1 site or 1000 sites: **one git repo**

Project Kepler

Kubernetes-based Efficient Power Level Exporter

eBPF + RAPL/hwmon + ML model → **per-pod, per-container, per-process watts**

-  CNCF **Sandbox** (since May 2023)
-  Latest stable: **v0.11.4** (Feb 2025)
-  github.com/sustainable-computing-io/kepler

2025 highlights:

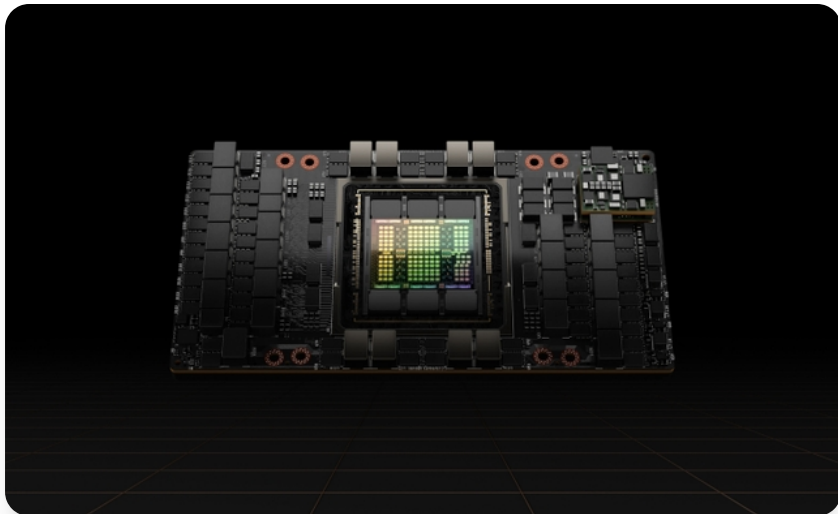
- GPU power (NVIDIA NVML)
- Redfish BMC fallback
- Kubelet polling rewrite
- Native Grafana Labs integration

Where the watts go — 2025 ballpark

Component	Idle	Load
 Server CPU (32c Xeon/EPYC)	~40 W	250–350 W
 Desktop CPU (i9 / Ryzen 9)	~15 W	125–250 W
 Raspberry Pi 5 / ARM SoC	~2 W	8–12 W
 32 GB DDR5 ECC (per stick)	~3 W	~6 W
 Consumer NVMe SSD	~1 W	5–8 W
 Enterprise NVMe (U.2)	~5 W	10–25 W
 Gaming GPU (RTX 4090)	~15 W	350–450 W
 AI GPU (NVIDIA H100 SXM)	~80 W	~700 W
 AI GPU (NVIDIA B200)	~150 W	~1000 W

⚡ One H100 ≈ **~100 Raspberry Pi 5s** flat out. An 8×H100 DGX node ≈ **~10.2 kW** — a small server room in one chassis.

NVIDIA H100 SXM



700 W

TDP at full load

~80 W

Idle

 **80 GB HBM3**

 **3.35 TB/s** memory bandwidth

 **3,958 TFLOPS FP16**

 **Released 2022 — Hopper architecture**

≈ 100× Raspberry Pi 5 power at load

NVIDIA B200



1,000 W

TDP at full load

~150 W

Idle

 **192 GB HBM3e**

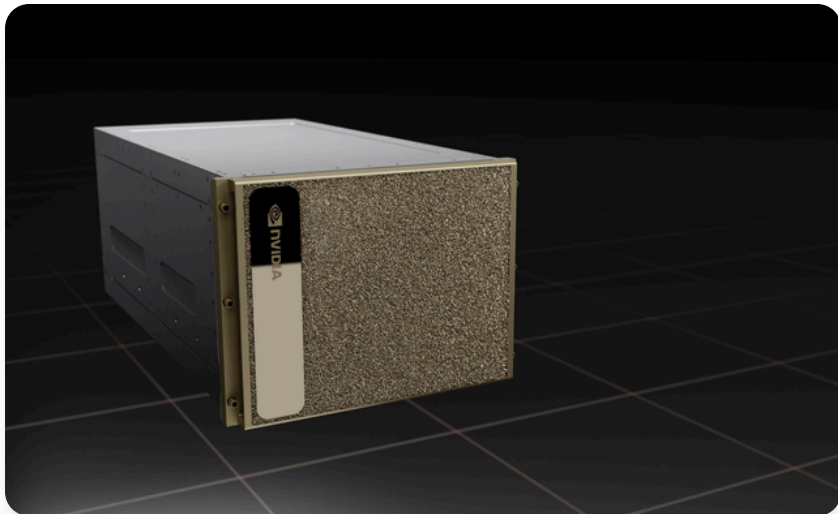
 **8 TB/s memory bandwidth**

 **~9 PFLOPS FP16**

 **Released 2024 — Blackwell architecture**

1.4× more power than H100 · 2.3× the bandwidth

NVIDIA DGX H100 — 8× H100 in one box 🏭



~10.2 kW

Full system TDP

- 🧠 **8× H100 SXM5 GPUs**
- 🧠 **640 GB total HBM3**
- 🔗 **NVLink 4.0** — 900 GB/s interconnect
- 💾 **2× 1.92 TB NVMe SSD**
- ⚡ **Requires 3-phase power supply**

A small server room in **one 6U chassis**.
~5× a typical rack's power budget — for a single node.

CNCF TAG Environmental Sustainability 🌱

🟢 Green Reviews

Public, per-project sustainability audits. Falco · KubeVirt · OpenTelemetry · ... **You maintain a CNCF project? Request one.**

📊 Cloud Native Sustainability Landscape

The map of who's doing what.

🎤 Recent talks worth watching

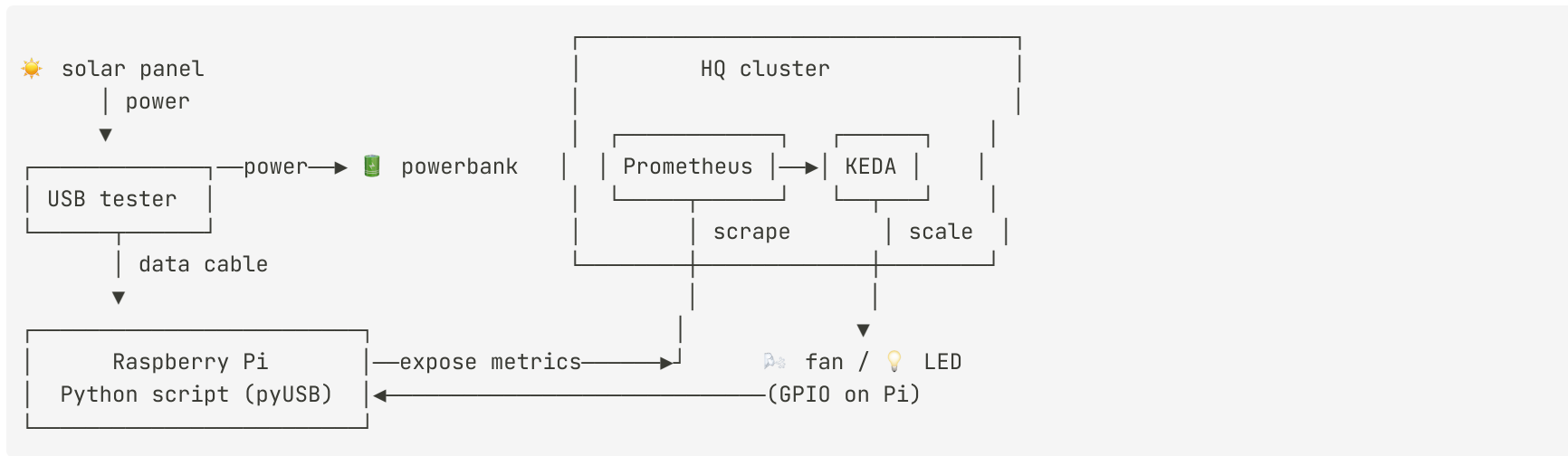
- *Green AI in Cloud Native* — KubeCon EU 2025
- *How Green is My OpenTelemetry Collector?* — KubeCon EU 2025
- *FinOps x GreenOps* — Sustainability Month Tokyo (Dec 2025)



`tag-env-sustainability.cncf.io` — weekly open meetings.

Demo time! 🌞💨

Carbon-aware HPA: the solar + fan demo 🌞🧠💡



USB tester measures solar watts → data cable to Pi → Python script reads USB tester & exposes metrics → HQ Prometheus scrapes → KEDA scales the fan / LED belt pod on the Pi.

Silly hardware, real loop. Same control flow on a real grid: **KEDA + carbon-aware-keda-operator** reads **WattTime / Electricity Maps** and time-shifts batch jobs to greener hours.

Refs: [Azure/carbon-aware-keda-operator](#) · [Kubernetes WG-Batch](#) · [GSF Carbon Aware SDK](#)

Key takeaways

1

K8s at the small edge is mature.

k0s + ArgoCD-from-HQ + shared Prometheus = a real fleet pattern, not a toy.

2

Power is observable now.

Kepler turns watts-per-pod from "trust the cloud bill" into a Prometheus metric you can alert on.

3

The next scaling axis is carbon.

HPA on CPU is table stakes. The win is scaling on grid intensity.

Same VM, different region: n4-standard-8 /year

8 vCPU / 32 GB · ~80 W avg load · ~700 kWh/yr · × Google's published per-region grid intensity (2024).



⚡ ~8.5x spread for the *same* machine. europe-west9 ≈ Paris flight; asia-south1 ≈ 10 Paris flights per VM/yr. Region selection is the cheapest single carbon decision your platform team can make.

Sources: cloud.google.com/sustainability/region-carbon (Google grid intensity 2024) · estimated 80 W avg, PUE 1.10.

Try it this week

1. Install Kepler in your existing cluster

```
helm repo add kepler \
  https://sustainable-computing-io.github.io/kepler-helm-chart
helm install kepler kepler/kepler -n kepler --create-namespace
```

2. Open your cloud's carbon dashboard

- ☁ GCP — Carbon Footprint console
- ☁ AWS — Customer Carbon Footprint Tool
- ☁ Azure — Emissions Impact Dashboard

3. Pick one CPU-family swap at your next FinOps review

Pinterest cut media-transcoding carbon **62%** by moving x86 → AWS Graviton.

Source: [AWS — Pinterest Graviton case study, 2024.](#)

Thank you 🌬️

Questions? Come *feel the breeze* on stage.

David Pech

github.com/depeche-io

Petr Rais

github.com/petrrais



github.com/depeche-io/kcd-2026

Repo + slides

Backup Video

If the Demo gods are not on our side